Midpoint of a segment

1

a $M\,(5,-2)$

b $M\,(10,6)$

2

a $M\,(5,7)$

b $M\left(\frac{-11}{2},-8\right)$

c $M\,(-3,2)$

d $M\,(-3,-2)$

e $M\,(-2,1)$

f $M\,(2,-4)$

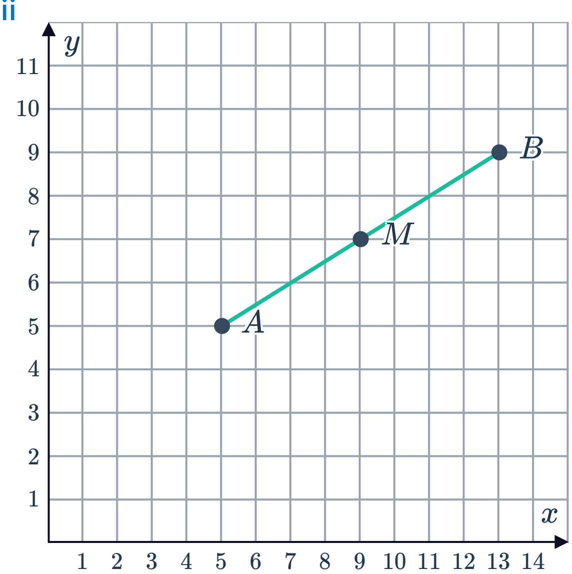
Midpoint formula (Extended)

3

a

i $M\,(9,7)$

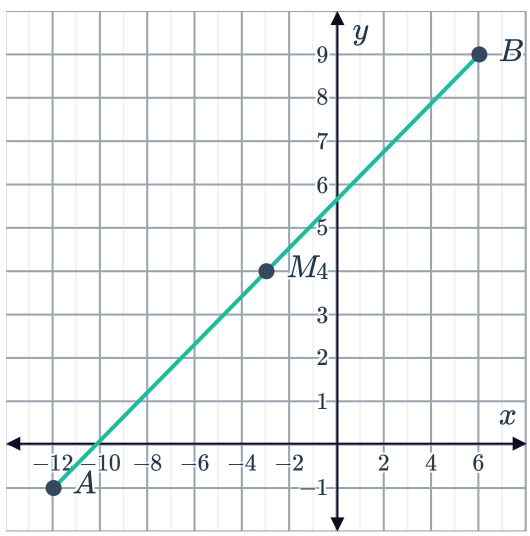
ii



b

i $M\,(-3,4)$

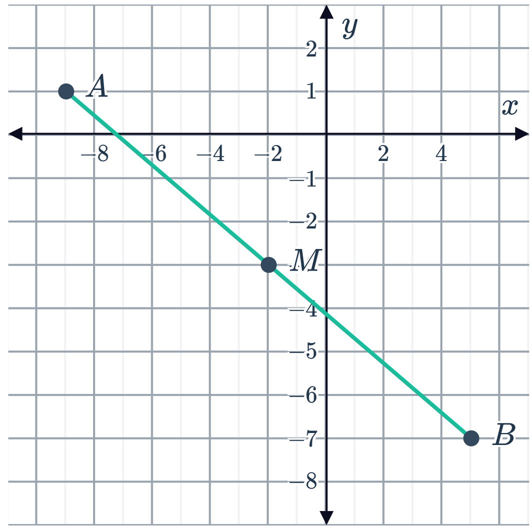
ii



c

i $M\,(-2,-3)$

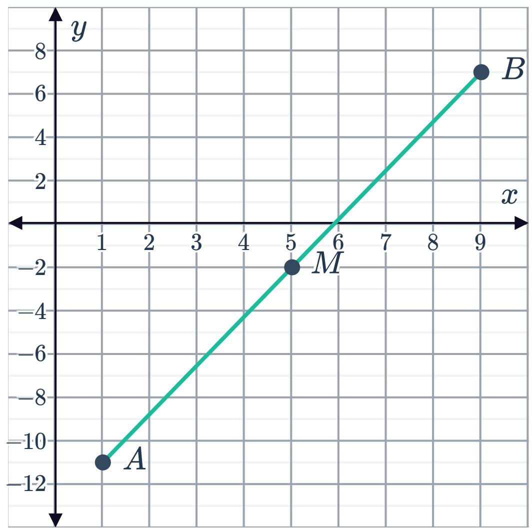
ii



d

i $M\,(5,-2)$

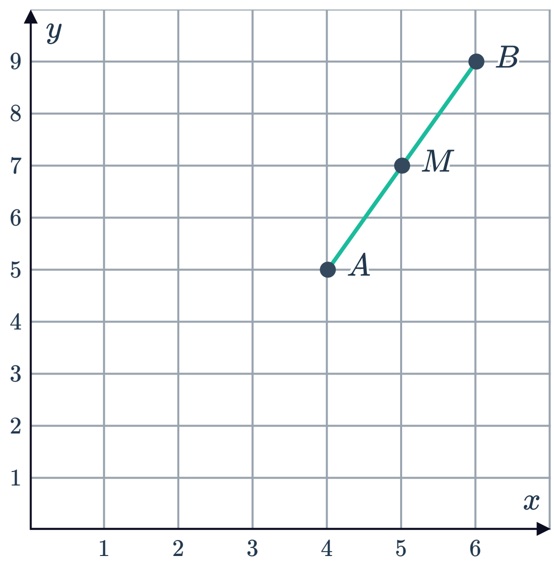
ii



e

i $M\,(5,7)$

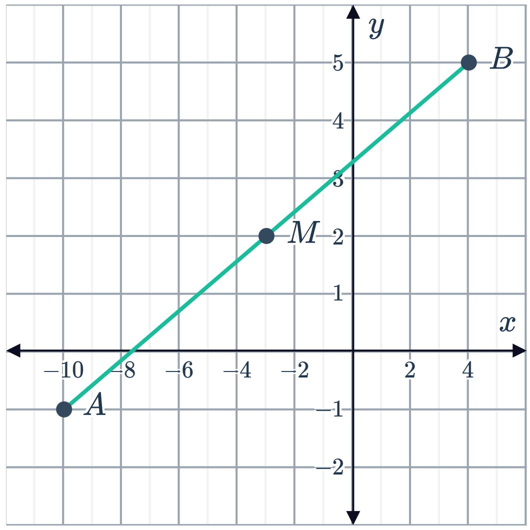
ii



f

i $M\,(-3, 2)$

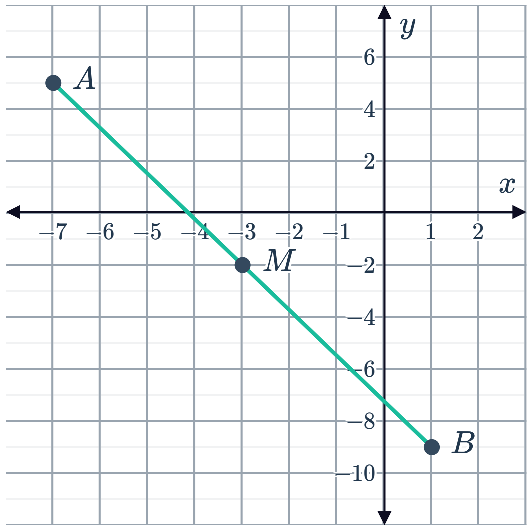
ii



g

i $M\,(-3, -2)$

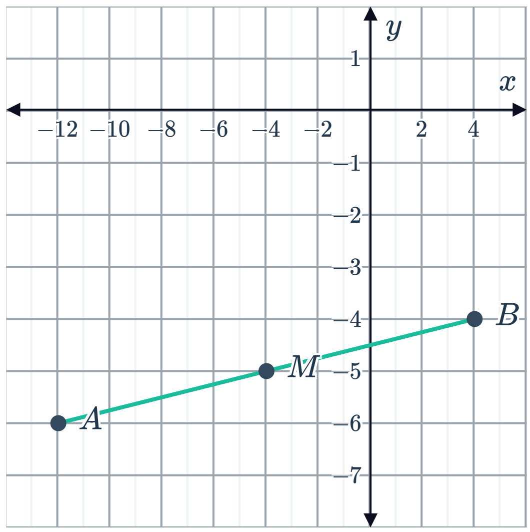
ii



h

i $M\,(-4, -5)$

ii



4

a $M\left(8,6\right)$

b $M\left(\frac{7}{2},\frac{1}{2}\right)$

c $M\left(-5,9\right)$

d $M\left(6,-9\right)$

e $M\left(-9,-4\right)$

f $M\left(-5,2\right)$

g $M\left(2,-6\right)$

h $M\left(-4,-7\right)$

i $M\left(-\frac{3}{4},\frac{3}{4}\right)$

j $M\left(3,\frac{5}{2}\right)$

5 $\left(4m,3n\right)$

6 $\left(m-3,\frac{31n}{12}\right)$

7

a $p=-\frac{1}{4}$

b $q=0$

8

a $M\left(3,7\right)$

b $k=-11$

9 $Q\left(-6,1\right),R\left(-8,4\right),S\left(-10,7\right)$

End points of a segment (Extended)

10

a $A\left(5,6\right)$

b $A\left(6,-5\right)$

c $A\left(5,6\right)$

d $A\left(5,-6\right)$

e $A\left(-5,6\right)$

f $A\left(4,-3\right)$

g $A\left(-4,-5\right)$

h $A\left(6,22\right)$

i $A\left(-3,3\right)$

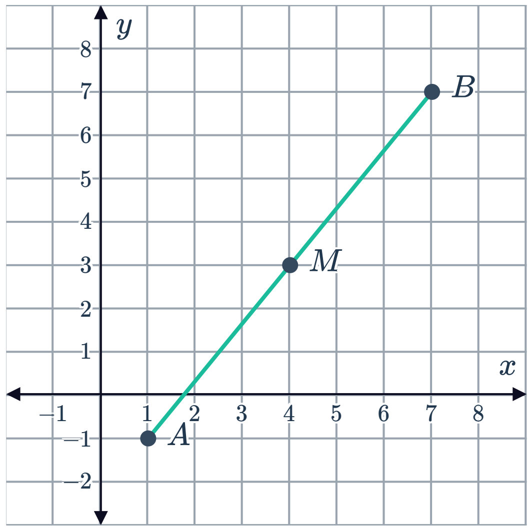
j $A\left(-7,-4\right)$

11

a

i $B\,(7,7)$

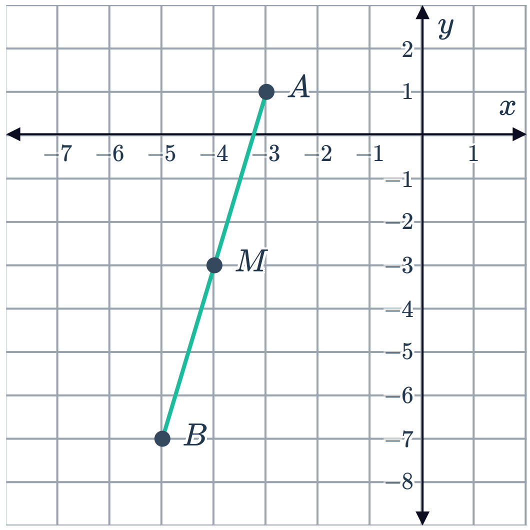
ii



b

i $B\,(-5,-7)$

ii



12 $B\,(8,-1)$

13

a Their diagonals bisect one another.

b

i $\left(-\dfrac{17}{2}, 3\right)$

ii $\left(-\dfrac{17}{2}, 3\right)$

c Yes

14

a $T\,(8,8)$

b $S\,(6,1)$

c Parallelogram, because the diagonals bisect each other.

Applications (Extended)

15 29 million

16 63 years

17 $\left(\frac{69}{2}{}^\circ\mathrm{N}, 103{}^\circ\mathrm{E}\right)$

18

a $207\,203$

b $268\,291$